Two-qubit Hamiltonian as example case

- We are given a general two-qubit Hamiltonian H, represented by a Hermitian 4×4 matrix.
- Goal: approximate the ground-state energy

$$E_0 = \min_{|\psi\rangle \neq 0} \frac{\langle\psi| H |\psi\rangle}{\langle\psi\rangle \psi} = \min_{|\psi\rangle,\ \|\psi\|=1} \langle\psi| H |\psi\rangle .$$

- We will use the **Variational Quantum Eigensolver (VQE)**:
  1. choose a parameterized state $|\psi(\boldsymbol{\theta})\rangle$ prepared by a quantum circuit (ansatz),
  2. estimate $E(\boldsymbol{\theta}) = \langle\psi(\boldsymbol{\theta})| H |\psi(\boldsymbol{\theta})\rangle$ from measurements,
  3. update parameters $\boldsymbol{\theta}$ on a classical optimizer to (approximately) minimize $E(\boldsymbol{\theta})$.

Measurement model: computational-basis (Pauli Z) readout

Most devices naturally measure each qubit in the computational basis $\{|0\rangle, |1\rangle\}$, i.e. eigenbasis of σ_z.

A single two-qubit measurement shot yields a bitstring $b_1 b_2 \in \{00, 01, 10, 11\}$. We map outcomes to eigenvalues:

$$\sigma_z |0\rangle = +1 |0\rangle, \qquad \sigma_z |1\rangle = -1 |1\rangle.$$

So define random variables per shot

$$z_1 = \begin{cases} +1, & b_1 = 0 \\ -1, & b_1 = 1 \end{cases}, \qquad z_2 = \begin{cases} +1, & b_2 = 0 \\ -1, & b_2 = 1 \end{cases}.$$

Then the estimator for $\langle \sigma_z \otimes \sigma_z \rangle$ is the sample mean of the product $z_1 z_2$. **Important:** Measuring *both* qubits in the Z basis gives access to $\sigma_z \otimes \sigma_z$ correlations in one shot.

Estimating $\langle Z_1 \rangle$, $\langle Z_2 \rangle$, $\langle Z_1 Z_2 \rangle$ from the same data

Let $p_{00}, p_{01}, p_{10}, p_{11}$ be probabilities of outcomes $|00\rangle, |01\rangle, |10\rangle, |11\rangle$. From Z-basis readout we have

$$\langle \sigma_z \otimes \mathbb{1} \rangle = (p_{00} + p_{01}) - (p_{10} + p_{11}),$$

$$\langle \mathbb{1} \otimes \sigma_z \rangle = (p_{00} + p_{10}) - (p_{01} + p_{11}),$$

$$\langle \sigma_z \otimes \sigma_z \rangle = (p_{00} + p_{11}) - (p_{01} + p_{10}).$$

Practical estimator: with N shots and counts n_{ij} (so $p_{ij} \approx n_{ij}/N$), plug in $p_{ij} = n_{ij}/N$ to obtain unbiased sample-mean estimators.

Why measuring both qubits is more convenient than only one

Suppose you only measure qubit 1 in Z and discard qubit 2. You can estimate $\langle \sigma_z \otimes \mathbb{1} \rangle$, but:

- You **cannot** directly estimate the two-qubit correlator $\langle \sigma_z \otimes \sigma_z \rangle$ without joint outcomes.
- Many Hamiltonians include interaction terms like $J\,\sigma_z \otimes \sigma_z$. Without joint readout, you would need additional circuitry or assumptions to reconstruct correlations.
- Joint measurement gives $\langle \sigma_z \otimes \mathbb{1} \rangle$, $\langle \mathbb{1} \otimes \sigma_z \rangle$, and $\langle \sigma_z \otimes \sigma_z \rangle$ **simultaneously** from the same shot record $\{b_1 b_2\}$.

Shot efficiency intuition: if one measurement record yields multiple observables, you reduce total shots needed to estimate the energy to a given precision.

Physics intuition: the ground-state energy of an interacting two-qubit system depends on *correlations*; you must measure both subsystems to access them.

Measuring general Pauli strings using Z-basis readout

Devices measure in Z, but VQE needs $\langle \sigma_x \otimes \sigma_x \rangle$, $\langle \sigma_y \otimes \sigma_z \rangle$, etc.

Strategy: rotate the measurement basis so that measuring Z after a unitary U is equivalent to measuring another Pauli operator before U.

Single-qubit identities:

$$H\sigma_z H = \sigma_x, \qquad S^\dagger H \sigma_z H S = \sigma_y,$$

where H is the Hadamard and $S = \mathrm{diag}(1, i)$.

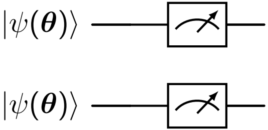
Thus,

- to measure σ_x on a qubit: apply H then measure in Z,
- to measure σ_y on a qubit: apply $S^\dagger$ then H then measure in Z,
- to measure σ_z: measure directly in Z.

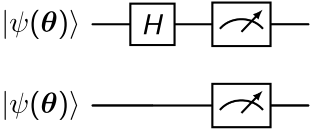
For two-qubit strings like $\sigma_x \otimes \sigma_y$, apply the appropriate basis-change gates to each qubit and then measure both in Z.

Quantikz: measurement circuits for Pauli strings

Example 1: measure $\sigma_z \otimes \sigma_z$ (no basis change)

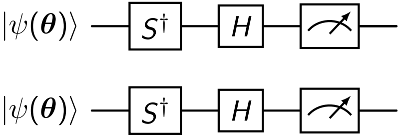


Example 2: measure $\sigma_x \otimes \sigma_z$ (Hadamard on qubit 1)



Quantikz: measurement circuits for Pauli strings

Example 3: measure $\sigma_y \otimes \sigma_y$ (apply $S^\dagger$ then H on each qubit)



Setup and claim

We consider a two-qubit state ρ and energy estimation for a Hamiltonian

$$H = \sum_{\alpha,\beta\in\{0,x,y,z\}} h_{\alpha\beta}\, \sigma_\alpha \otimes \sigma_\beta, \qquad \sigma_0 = \mathbb{1}.$$

Claim (informal). Measuring only the first qubit—even after arbitrary pre-rotations—cannot, in general, determine two-qubit correlators such as $\langle \sigma_z \otimes \sigma_z \rangle$, hence cannot determine $\langle H \rangle$ for generic interacting Hamiltonians.

We will:

1. prove formally that any single-qubit measurement depends only on the reduced state $\rho_1 = \mathrm{Tr}_2\, \rho$,
2. exhibit distinct states with identical ρ_1 but different $\langle \sigma_z \otimes \sigma_z \rangle$,
3. give a worked VQE example where relying on single-qubit measurements provably fails.

Formal fact 1: single-qubit statistics depend only on the reduced state

Let $\{M_m\}$ be a POVM performed *only* on qubit 1. Operationally this corresponds to measurement operators $\{M_m \otimes \mathbb{1}\}$ on the two-qubit system.

The probability of outcome m is

$$p(m) = \mathrm{Tr}\big[(M_m \otimes \mathbb{1})\rho\big].$$

Using the defining property of the partial trace,

$$\mathrm{Tr}\big[(A \otimes \mathbb{1})\rho\big] = \mathrm{Tr}\big[A\ \mathrm{Tr}_2(\rho)\big],$$

we obtain

$$p(m) = \mathrm{Tr}\big[M_m\,\rho_1\big], \qquad \rho_1 := \mathrm{Tr}_2(\rho).$$

Conclusion. *All* statistics from measurements that act only on qubit 1 depend only on ρ_1. They are completely insensitive to correlations that live in ρ but vanish under Tr_2.

Corollary: no single-qubit readout can determine two-qubit correlators

A two-qubit correlator (e.g. $\langle \sigma_z \otimes \sigma_z \rangle$) is

$$\langle \sigma_z \otimes \sigma_z \rangle_\rho = \mathrm{Tr}\big[(\sigma_z \otimes \sigma_z)\rho\big].$$

From the previous slide, any single-qubit protocol (measurements on qubit 1 only) can at best determine ρ_1. Therefore, if there exist ρ, ρ' such that

$$\mathrm{Tr}_2(\rho) = \mathrm{Tr}_2(\rho') \quad \text{but} \quad \mathrm{Tr}\big[(\sigma_z \otimes \sigma_z)\rho\big] \neq \mathrm{Tr}\big[(\sigma_z \otimes \sigma_z)\rho'\big],$$

then $\langle \sigma_z \otimes \sigma_z \rangle$ cannot be inferred from single-qubit measurement data.

Next slide: we construct such a pair explicitly (pure states, maximally entangled).

Exhibit: same reduced state, different $\langle Z \otimes Z \rangle$

Consider the Bell states

$$|\Phi^+\rangle = \frac{|00\rangle + |11\rangle}{\sqrt{2}}, \qquad |\Psi^+\rangle = \frac{|01\rangle + |10\rangle}{\sqrt{2}}.$$

Let $\rho_{\Phi^+} = |\Phi^+\rangle\langle\Phi^+|$ and $\rho_{\Psi^+} = |\Psi^+\rangle\langle\Psi^+|$.

Compute reduced states. For either state,

$$\rho_1 = \mathrm{Tr}_2(\rho) = \frac{\mathbb{1}}{2}.$$

Hence *every* measurement on qubit 1 alone yields identical statistics for $|\Phi^+\rangle$ and $|\Psi^+\rangle$.

Compute the correlation. Using $\sigma_z |0\rangle = +|0\rangle$, $\sigma_z |1\rangle = -|1\rangle$,

$$(\sigma_z \otimes \sigma_z)|00\rangle = +|00\rangle, \quad (\sigma_z \otimes \sigma_z)|11\rangle = +|11\rangle,$$
$$(\sigma_z \otimes \sigma_z)|01\rangle = -|01\rangle, \quad (\sigma_z \otimes \sigma_z)|10\rangle = -|10\rangle.$$

Therefore,

$$\langle \sigma_z \otimes \sigma_z \rangle_{\Phi^+} = +1, \qquad \langle \sigma_z \otimes \sigma_z \rangle_{\Psi^+} = -1.$$

Formal theorem: impossibility of recovering generic energies from one-qubit measurements

Theorem. Let H be any Hamiltonian that contains at least one nonzero two-qubit term, e.g. $h_{zz}\,\sigma_z \otimes \sigma_z$ with $h_{zz} \neq 0$. Then there exist states ρ, ρ' such that:

$$\mathrm{Tr}_2(\rho) = \mathrm{Tr}_2(\rho') \quad \text{but} \quad \mathrm{Tr}(H\rho) \neq \mathrm{Tr}\big(H\rho'\big).$$

Hence the energy $\langle H \rangle$ cannot be determined from measurements on qubit 1 alone.

Proof (constructive). Take $\rho = \rho_{\Phi^+}$, $\rho' = \rho_{\Psi^+}$. Then $\mathrm{Tr}_2(\rho) = \mathrm{Tr}_2(\rho') = \mathbb{1}/2$, but for the h_{zz} term,

$$\mathrm{Tr}\big[(\sigma_z \otimes \sigma_z)\rho\big] = +1, \qquad \mathrm{Tr}\big[(\sigma_z \otimes \sigma_z)\rho'\big] = -1.$$

Therefore

$$\mathrm{Tr}(H\rho) - \mathrm{Tr}\big(H\rho'\big) \supset h_{zz}\big(+1 - (-1)\big) = 2h_{zz} \neq 0,$$

so $\mathrm{Tr}(H\rho) \neq \mathrm{Tr}(H\rho')$, while all one-qubit statistics on qubit 1 are identical. $\square$

Addressing the "pre-rotation" idea: why it doesn't save single-qubit readout

One may attempt: apply a two-qubit unitary U, then measure only qubit 1 in the Z basis. This measures the observable

$$A := U^\dagger(\sigma_z \otimes \mathbb{1})U$$

on the original state.

Two key limitations:

- **Single observable per circuit.** Measuring only qubit 1 gives information about expectation values of *one* operator A per measurement setting. VQE generally needs many Pauli strings $\{\sigma_\alpha \otimes \sigma_\beta\}$.
- **No universal compression.** There is no fixed unitary U such that $U^\dagger(\sigma_z \otimes \mathbb{1})U$ equals $\sigma_\alpha \otimes \sigma_\beta$ for *all* required terms simultaneously. You would need a different U per term anyway, eliminating any advantage over simply measuring both qubits.

Conceptual punchline: two-qubit Hamiltonians depend on correlations; correlations are not encoded in a single reduced state.

Worked VQE failure example: Hamiltonian requiring correlations

Consider the interacting Hamiltonian

$$H = -\,\sigma_z \otimes \sigma_z.$$

Its eigenvalues are $\{-1, -1, +1, +1\}$ with ground energy $E_0 = -1$. Any ground state must satisfy $\langle \sigma_z \otimes \sigma_z \rangle = +1$ (e.g. $|00\rangle$, $|11\rangle$, or any superposition in the $+1$ eigenspace).

Energy functional:

$$E(\rho) = \mathrm{Tr}(H\rho) = -\,\langle \sigma_z \otimes \sigma_z \rangle_\rho.$$

Now compare the two Bell states from earlier:

$$E(\Phi^+) = -\,(+1) = -1, \qquad E(\Psi^+) = -\,(-1) = +1.$$

They differ by $\Delta E = 2$, but have identical $\rho_1 = \mathbb{1}/2$.

Therefore any VQE evaluation strategy that uses only qubit-1 measurement data *cannot even tell whether the energy is -1 or $+1$.*

VQE ansatz family demonstrating the failure concretely

Take a simple two-qubit ansatz capable of preparing both Bell states:

$$|\psi(\theta)\rangle = U(\theta)\,|00\rangle\,, \quad U(\theta) := \mathrm{CNOT}_{1\to 2}\,(R_y(\theta)\otimes \mathbb{1}).$$

One checks:

$$\theta = \frac{\pi}{2} \;\Rightarrow\; |\psi(\theta)\rangle = |\Phi^+\rangle\,, \qquad \theta = \frac{\pi}{2} \text{ with an extra } X \text{ on qubit } 2 \Rightarrow |\Psi^+\rangle\,.$$

But for both $|\Phi^+\rangle$ and $|\Psi^+\rangle$,

$$\rho_1 = \mathrm{Tr}_2(|\psi\rangle\langle\psi|) = \frac{\mathbb{1}}{2} \quad \Rightarrow \quad \langle \sigma_z \otimes \mathbb{1}\rangle = \mathrm{Tr}(\sigma_z \rho_1) = 0,$$

and similarly any single-qubit observable on qubit 1 has the same expectation (because $\rho_1 = \mathbb{1}/2$).

Thus: if your VQE "measurement layer" reads only qubit 1, the cost function becomes *constant* across this family and provides no gradient/no signal for optimization.

What measurement on both qubits provides (and why it fixes VQE)

If you measure *both* qubits in the computational basis, each shot yields $b_1 b_2 \in \{00, 01, 10, 11\}$. Map bits to ± 1 eigenvalues:

$$z_1 = (-1)^{b_1}, \qquad z_2 = (-1)^{b_2}.$$

Then a single shot provides:

$$z_1 \Rightarrow \sigma_z \otimes \mathbb{1}, \qquad z_2 \Rightarrow \mathbb{1} \otimes \sigma_z, \qquad z_1 z_2 \Rightarrow \sigma_z \otimes \sigma_z.$$

So the estimator for $\langle \sigma_z \otimes \sigma_z \rangle$ is

$$\langle \widehat{\sigma_z \otimes \sigma_z} \rangle = \frac{1}{N} \sum_{s=1}^{N} z_1^{(s)} z_2^{(s)},$$

and the energy estimate is

$$\hat{E} = -\langle \widehat{\sigma_z \otimes \sigma_z} \rangle.$$

Efficiency point: the same shot record gives multiple observables at once; measuring only one qubit throws away correlation information and wastes shots.

Generalization: why this is not a special case

A generic two-qubit Hamiltonian has the Pauli expansion

$$H = \sum_{\alpha,\beta} h_{\alpha\beta}\, \sigma_\alpha \otimes \sigma_\beta,$$

with up to 15 nontrivial coefficients $(\alpha, \beta) \neq (0, 0)$.

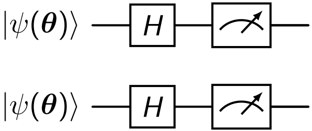
If H contains any interaction term (e.g. $\sigma_z \otimes \sigma_z$, $\sigma_x \otimes \sigma_x$, $\sigma_y \otimes \sigma_y$, $\sigma_x \otimes \sigma_z$, etc.), then the energy depends on correlators $\langle \sigma_\alpha \otimes \sigma_\beta \rangle$.

But measuring only qubit 1 gives access only to $\langle \sigma_\alpha \otimes \mathbb{1} \rangle$, i.e. local expectations.

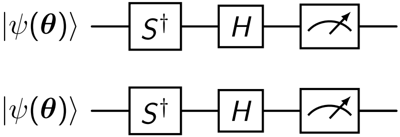
Formal obstruction: There exist inequivalent two-qubit states sharing the same ρ_1, so no function of ρ_1 alone can reproduce all correlators. Thus single-qubit measurement cannot generally yield the correct energy landscape for VQE.

More examples of measurement circuits

Modern devices measure all qubits in the Z basis; other Pauli strings are measured via pre-rotations. **(i)** $X \otimes X$**:** apply H on both, then Z readout



(ii) $Y \otimes Y$**:** apply $S^\dagger$ then H on both, then Z readout



Why joint Z readout is sufficient for energy estimation

Because each Pauli string measurement is reduced to:

1. local basis-change gates $U_k = U_k^{(1)} \otimes U_k^{(2)}$,
2. joint Z readout producing bitstrings $b_1 b_2$,

we have

$$\langle P_k \rangle_\psi = \langle (\sigma_z^{p_1} \otimes \sigma_z^{p_2}) \rangle_{U_k \psi} \quad \text{with} \quad p_i \in \{0, 1\},$$

where the mapping depends on whether the rotated measurement corresponds to $\sigma_x, \sigma_y, \sigma_z$.
Operationally: you always measure in the computational basis, but you interpret the bits as ± 1 eigenvalues (after the appropriate pre-rotations).
And: joint readout gives you correlators (e.g. $Z_1 Z_2$) needed for interaction energies.

(Optional) Operator grouping and measurement efficiency

Many Pauli strings $\{P_k\}$ commute and can be measured in the same basis. For two qubits, an important commuting set is:

$$\{\sigma_z \otimes \mathbb{1},\ \mathbb{1} \otimes \sigma_z,\ \sigma_z \otimes \sigma_z\}$$

which all share the Z basis and are estimated from the *same* Z-basis shots.

Similarly, $\{\sigma_x \otimes \sigma_x,\ \sigma_x \otimes \mathbb{1},\ \mathbb{1} \otimes \sigma_x\}$ share the X basis (apply H on relevant qubits), and analogous for Y basis.

Benefit: fewer distinct measurement circuits $\Rightarrow$ fewer total shots for a target energy precision.

Summary

- Any two-qubit Hermitian 4×4 Hamiltonian admits a Pauli expansion $H = \sum h_{\alpha\beta}\sigma_\alpha \otimes \sigma_\beta$.
- VQE minimizes $E(\boldsymbol{\theta}) = \langle\psi(\boldsymbol{\theta})| H |\psi(\boldsymbol{\theta})\rangle$ using the variational principle.
- Energy estimation reduces to estimating expectation values of Pauli strings.
- All measurements can be performed with **computational-basis (Z)** readout by inserting local basis-change gates.
- Measuring **both qubits** is more convenient because it yields $\sigma_z \otimes \sigma_z$ (and other two-qubit correlators) directly and improves shot efficiency by extracting multiple observables from one record.

More Summary (what you should remember)

- **Formal fact:** any measurement acting only on qubit 1 depends only on the reduced state $\rho_1 = \mathrm{Tr}_2\, \rho$.
- **Correlations are invisible locally:** there exist states with the same ρ_1 but different $\langle \sigma_\alpha \otimes \sigma_\beta \rangle$.
- **VQE requires correlators** because interacting Hamiltonians include two-qubit terms at least.
- **Practical takeaway:** measuring both qubits in the Z basis yields local expectations and correlators from the same bitstrings, enabling correct and shot-efficient energy estimation.